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Branch: Industrial Chemistry

39/50

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1. (d) The asthenosphere is present just below the

- (a) Stratosphere
(c) Continental Crust

- (b) Ionosphere
(d) Upper mantle

2. (a) What is the name of the instrument used for measurements of ozone concentrations?

- (a) Long Beam UV-Photometer
(c) Dual Beam UV-Photometer

- (b) Narrow Beam UV-Photometer
(d) Large cell UV-Photometer

3. (d) Which oxidizing agent is known as the detergent of the atmosphere?

- (a) NO
(c) OOH radical

- (b) H_2O_2
(d) OH radical

4. (c) Which of the following is not among the five spheres of the environment?

- (a) Atmosphere
(c) Lakosphere

- (b) Biosphere
(d) Anthrosphere

5. (d) The Rh catalyst in the three-catalytic converter is used for

- (a) production of H_2
(c) reduction of hydrocarbons

- (b) production of NO
(d) oxidation of CO

6. (c) Where is the absorption maximum located in the absorption spectrum of ozone in the 200 to 350 nm wavelength range?

- (a) 200-210 nm
(c) 250-260 nm

- (b) 220-230 nm
(d) 270-280 nm

7. (c) Which of the parameters of Lambert-Beer's law is made very large for in-situ measurement of ozone in the atmosphere?

- (a) Concentration
(c) path length

- (b) Molar extinction coefficient
(d) Molar absorptivity

8. (b) How many nodes are there in the $\sigma_u^*(2p_z)$ of the CO molecule?

- (a) None
(c) Two

- (b) One
(d) Three

9. (b) What is the B.O. of 1O_2 ($^1\Delta$)?

- (a) one
(c) three

- (b) two
(d) none of the above

10. The stretching frequency of bridging carbonyl is ----- than that of the terminal carbonyl in the metal carbonyl compounds.

- (a) less
(c) equal

- (b) more
(d) none of the above

11. Among the following metal carbonyls, which has the lowest value of stretching frequency of CO bond?

- (a) $Cr(CO)_6$
(c) $[Ti(CO)_6]^{2+}$

- (b) $Co_2(CO)_8$
(d) $Mn_2(CO)_{10}$

12. Deoxy hemoglobin is

- (a) six coordinated
(c) four coordinated

- (b) five coordinated
(d) three coordinated

a is the answer

13. The binding affinity of the free Heme group for CO is about _____ times than O₂
(b) ☒ (a) 2500 (b) 0.25 lakh
(c) 2.5 lakh (d) 25 lakh

14. (b) Which of the following is an example of green pesticides
(a) Spinosed ☒ (b) Spinosad
(c) Spinosat (d) None of the above

15. (a) Which of the plants is badly affected by N₂O gas?
☒ (a) Alfalfa (b) Rice
(c) Lotus (d) None of the above

16. (a) The typical pressure used in the Born Haber process is
☒ (a) 200 atm (b) 2 atm
(b) 20 atm (d) 2000 atm

17. (c) Which of the NO_x is colored at room temperature?
☒ (a) NO (b) N₂O₅
☒ (c) NO₂ (d) N₂O

18. (d) How many hydroxy radicals are generated by the photocatalytic reaction of RCH₃ in photochemical smog
☒ (a) 10,000 (b) 100
(c) 10 ☒ (d) 4

19. (a) The role of Rh in the two-way catalytic converter is to convert
☒ (a) NO to O₂ (b) None of the above
(c) HC to CO₂ (d) HC to H₂

20. (c) What is the chemical name of H₂S₂O₄?
☒ (a) Thiosulphuric acid (b) Dithionic acid
☒ (c) Dithionous acid (d) None of the above

21. (c) Which chemical was developed as a pesticide for cotton industries in the USA?
☒ (a) Endrin (b) DDT
☒ (c) Toxaphene (d) None of the above

22. (c) Which chemical is majorly responsible for the eye irritation of the photochemical smog?
☒ (a) N₂O₅ (b) NO
☒ (c) NO₂ (d) H₃C-(CO)-O-O-NO₂ (PAN)

23. (b) One Dobson unit is equal to
☒ (a) 2.7 x 10²³ ozone/sq centimeter ☒ (b) 2.7 x 10²⁰ ozone/sq meter
(c) 2.7 x 10¹⁶ ozone/sq meter (d) 2.7 x 10²⁰ ozone/sq centimeter

24. (d) Which Cr ion coming from leather tanneries can be precipitated as a solid?
☒ (a) Cr(III) (b) Cr(II)
(c) Cr(IV) ☒ (d) Cr(VI)

25. (c) The chemical formula of Toxaphene is
☒ (a) C₁₀H₁₀Cl₈ (b) C₁₂H₁₂Cl₉
☒ (c) C₁₀H₁₀Cl₉ (d) C₁₀H₁₀Cl₁₀

26. (a) Enzyme is a
☒ (a) Biocatalyst (b) Non-protein material (c) Halocarbon (d) None of the options.

27. The active metal center of particulate methane monooxygenase (p-MMO) is

- (a) Fe (b) ☒ Cu (c) Mo (d) Co

28. Which of the following molecules is an example of PAH?

- (a) Butane (b) ☒ Naphthalene (c) Methane (d) Propane

29. Which one of the following is used as a growth substrate for *Pseudomonas methanica*?

- (a) ☒ Methane (b) Ethane (c) Propane (d) TCE

30. The slope of the Lineweaver-Burk plot for enzyme-substrate reaction kinetics is

- (a) v_{max} (b) $\frac{1}{v_{max}}$ (c) ☒ $\frac{K_M}{v_{max}}$ (d) $\frac{1}{S_0}$

The terms denote their conventional significances.

31. The intercept of the Lineweaver-Burk plot for enzymatic kinetics is

- (a) v_{max} (b) ☒ $\frac{1}{v_{max}}$ (c) $\frac{K_M}{v_{max}}$ (d) $\frac{1}{S_0}$

32. In the presence of an inhibitor in the enzyme-substrate reaction, which of the following is correct for non-competitive enzyme inhibition?

- (a) Inhibitor binds to an enzyme and reduces its capability to bind with substrate
(b) ☒ Inhibitor binds to enzyme and enzyme-substrate complex both
(c) Inhibitor binds only with the enzyme
(d) Inhibitor binds only with the enzyme-substrate complex

33. Krebs's cycle is

- (a) 4-step process (b) ☒ 8-step process (c) 6-step process (d) 10-step process

34. In Krebs's cycle, which of the following enzymes transform citrate into isocitrate?

- (a) ☒ α -ketoglutarate (b) Aconitase (c) Isocitrate (d) Fumarase

35. Environmental factor (E-factor) is the measurement of

- (a) ☒ Total Waste (kg) / Product (kg)
(b) Product (kg) / Total Waste (kg)
(c) Total Waste (kg) / Total reactant (kg)
(d) Reactant (kg) / Product (kg)

36. The Atom Economy concept was developed by

- (a) ☒ Prof. B.M Trost (b) Prof. Hoechst (c) Prof. Paul T. Anastas (d) Prof. John C. Warner

37. In 2002, Nature Works, a Cargill company, won a Green Chemistry award for producing

- (a) ☒ Biodegradable plastic from corn
(b) Atom economy
(c) Industrial ecology (d) Ibuprofen production

38. FAD is

- (a) ☒ Flavin adenine dinucleotide
(b) Flavin adenine trinucleotide
(c) Flavin adenosine dinucleotide
(d) ☒ Flavin adenosine dinucleotide hydrate

39. In β -oxidation of fatty acid molecules, which one of the following is produced as the final product?

- (d) ☒ (a) Fatty acyl CoA (b) β -Ketoacyl CoA ☒ (c) Acetyl-CoA ☒ (d) β -Hydroxyacyl CoA

40. Which one among the following compounds decomposes at the slowest rate?

- ☒ (a) Lignin (b) Sugar (c) Protein (d) Starch

41. PCBs are

- (b) ☒ (a) Pyroloquinoline benzenes (c) Polychlorinated butanes ☒ (d) Polychlorinated biphenyls ☒ (e) Polychlorinated butyl ethers.

42. In Candida Maltose the conversion of alkane to 1-Fatty Alcohol is carried out by the following enzyme

- (c) ☒ (a) Fatty alcohol oxidase (b) Fatty aldehyde reductase ☒ (c) P450 enzyme (d) FAD

43. Proteins are the polymer of

- (b) ☒ (a) Sugars ☒ (b) Amino acid (c) Carboxylic acids (d) Aliphatic amines

44. The synthesis of catechol from glucose using biocatalytic has been demonstrated using the following bacteria

- ☒ (a) E. Coli (b) Pseudomonas Putida (c) Candida maltose (d) Candida tropicalis

45. Ionic liquids are considered the ideal green solvents because they are

- ☒ (a) Non-volatile, non-flammable and non-explosive (b) Unstable at high temperatures
(c) Volatile at room temperature (d) Explosive

46. Which is not the characteristic of a consumable product?

- (d) ☒ (a) Degradable (b) Non-bio-accumulative (c) Non-toxic ☒ (d) Recyclable

47. Biodegradation of long-chain alkane present in petroleum follows the following biodegradation pathways.

- ☒ (a) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acid $\rightarrow$ Acetyl CoA
(b) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acetyl CoA $\rightarrow$ Acid
(c) Alkane $\rightarrow$ Acetyl CoA $\rightarrow$ Aldehyde $\rightarrow$ Alcohol $\rightarrow$ Acid
(d) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acetyl CoA $\rightarrow$ Acid

48. Industrial ecology implies

- (d) ☒ (a) Industrial waste or by-products become significant input for another process.
(b) Recycling each other's waste materials
(c) Efficient utilization of energy
☒ (d) All the options are correct

49. Which of the following is one of the most notable examples of Industrial Ecology in practice today?

- ☒ (a) The Hoover Dam ☒ (b) The Redwood Forest in California
☒ (c) Denmark's Kalundborg Eco-industrial Par (d) The solar power plants in the Mojave Desert

50. TCE is

- ☒ (a) Trichloroethylene (b) Trichloroethane (c) Trichloroethanol (d) Trichloroepoxide